

Tribhuvan University
Institute of Science and Technology



Project Proposal

On

“Retail Management System”

A project proposal submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science in Computer Science and Information Technology.

(B.SC. CSIT)

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1. Introduction

The rapid growth of digital technology has significantly transformed the retail industry, shifting traditional business operations toward online and web-based platforms. E-commerce systems have become essential tools for managing products, customers, and transactions efficiently while providing convenience and accessibility to users. As consumers increasingly prefer online shopping, there is a strong need for reliable retail management systems that can support day-to-day retail operations in a secure, organized, and user-friendly manner.

A **Retail Management System** is designed to help retailers manage their online business activities such as product listing, customer registration, shopping cart management, order processing, and payment handling through a centralized web application. By automating these processes, the system reduces manual effort, minimizes errors, and improves overall operational efficiency. At the same time, customers benefit from easy product browsing, smooth checkout processes, and personalized account management.[1]

The proposed project focuses on developing a web-based e-commerce Retail Management System using modern web technologies. The system aims to provide essential functionalities including user authentication, product and category management, cart and order handling, payment integration, and email-based notifications. The application is structured using a modular approach to ensure maintainability, scalability, and clear separation of responsibilities across different components of the system.

In addition to standard e-commerce functionalities, the system also includes an image-based product search feature. This feature allows users to search for products by uploading an image instead of typing product names. The system analyzes basic visual characteristics such as color, shape, and patterns to find and display similar products available in the store. This functionality improves product discovery, especially for users who may not know the exact name or description of a product. [2]

This project demonstrates the practical application of web development concepts, database management, and software engineering principles along with the integration of ML features in web applications.

2. Problem Statement

The rapid expansion of online shopping has transformed customer expectations, with users increasingly demanding fast, intelligent, and visually driven product search experiences; however, many retail systems used by small and medium-sized businesses still rely solely on text-based search, creating challenges when customers wish to search using images such as photos seen on social media, in magazines, or captured personally without knowing the exact product name or description. Additionally, many existing retail platforms operate through disconnected systems where inventory, orders, payments, customer accounts, and notifications are managed separately or manually, resulting in data inconsistencies, slow processing, duplicate data entry, and poor coordination between front-end user interfaces and back-end operations. Furthermore, most current systems lack advanced features such as real-time error detection, user behavior analytics, secure payment integration, and personalized shopping experiences, which limits retailers' ability to resolve issues promptly, understand customer preferences, reduce cart abandonment, and build customer trust; therefore, there is a strong need for a unified, intelligent, and scalable web-based retail management system that addresses these challenges in an integrated manner.

3. Objectives

The project aims to develop a modern, intelligent, and user-friendly web-based Retail Management System that enhances online shopping through AI-powered visual search and complete retail operation features.

The major objectives of the Retail Management System project are as follows:

1. To create a centralized web platform that allows customers to browse products, manage shopping carts, place orders, make secure payments, receive email notifications, and handle personal accounts.
2. To implement visual search functionality so users can upload product images to find matching or similar items, making product discovery faster and more intuitive.
3. To gain practical experience in full-stack web development and ML integration.

4. Methodology

This project will be developed using the **Agile software development methodology**. Agile is selected because it supports flexibility, iterative development, and continuous evaluation, which are essential for building modern web-based applications. Since project requirements may evolve during development, the Agile approach allows improvements and refinements to be incorporated efficiently throughout the development lifecycle.

The development process will be carried out in a series of short and manageable iterations known as sprints. Each sprint will involve planning, design, implementation, and testing activities for a set of system functionalities. At the end of every sprint, the progress will be reviewed to evaluate system performance and identify areas for improvement before proceeding to the next iteration.

Testing and evaluation will be conducted continuously during the development process to ensure functionality, reliability, and usability of the system. Feedback obtained during testing phases will be used to enhance subsequent development cycles. By following the Agile methodology, the project aims to ensure structured progress, reduced development risk, and the delivery of a reliable and user-friendly Retail Management System that meets the defined objectives of the proposal.



Fig: Agile Methodology [3]

4.1 Requirement Identification

4.1.1 Literature Review

Recent research highlights the growing role of artificial intelligence in transforming ecommerce and retail management. Studies show that traditional text-based search methods in online retail often limit product discovery, especially when users lack precise keywords or product names. AI-powered **visual search**, using deep learning and image recognition, addresses this by allowing users to upload images to find similar products based on visual features like color, shape, pattern, and style. For instance, research on visual search engines for e-retail demonstrates how convolutional neural networks (CNNs) and computer vision techniques enable more accurate and intuitive product matching, significantly improving user satisfaction and reducing search time.[4]

Several works emphasize the benefits of visual search in e-commerce platforms. Papers and industry analyses indicate that AI-driven visual search increases engagement, reduces cart abandonment, and boosts conversion rates by up to 30% in some cases, as seen in implementations by major retailers like Amazon (StyleSnap), ASOS, and Pinterest Lens. Deep learning models, such as ResNet-based embeddings and object detection algorithms, are commonly used to extract and compare image features, making visual search effective even for complex categories like fashion, furniture, and home decor. Additionally, studies on AI in retail discuss broader integrations, including personalized recommendations, inventory optimization, and real-time analytics, which enhance overall system efficiency and customer experience.

Existing literature also covers comprehensive retail management systems, often focusing on secure authentication, order processing, payment handling, and basic analytics. However, few studies combine end-to-end retail operations with AI-based visual search in a single scalable web platform, particularly for small to medium retailers. Most prior work remains conceptual or limited to specific features, with gaps in practical integration of monitoring tools and user behavior analysis. The proposed Retail Management System aims to fill this gap by developing a unified, web-based solution using Agile methodology, incorporating AI visual search alongside standard e-commerce features for improved usability, reliability, and real-world applicability.

4.1.2 Requirement Analysis

Requirement analysis is a crucial phase of the project proposal, as it identifies and defines the expectations and constraints of the system before development begins. The requirement analysis for the proposed Retail Management System focuses on understanding what the system is expected to do (functional requirements) and how well it should perform under various conditions (non-functional requirements). This analysis helps ensure that the system design and development align with user needs and project objectives.

Functional Requirements

Functional requirements specify the essential services the system will provide.

1. User Account Management

The system will allow users to create, access, and manage their accounts securely.

2. Role-Based Access Control

Different user roles will be supported to ensure controlled and secure access to system features.

3. Product Browsing and Viewing

Users will be able to view and browse products through an organized and structured interface.

4. Online Shopping and Order Processing

The system will support online purchasing and systematic handling of order information.

5. Payment Handling

The system will facilitate secure processing of online payments.

6. Image-Based Product Search

Users will be able to upload an image to search for visually similar products.

7. E-mail Based Notification

The system will provide notifications to inform users about important system activities.

8. Administrative Management

Administrative users will be able to oversee and manage system operations

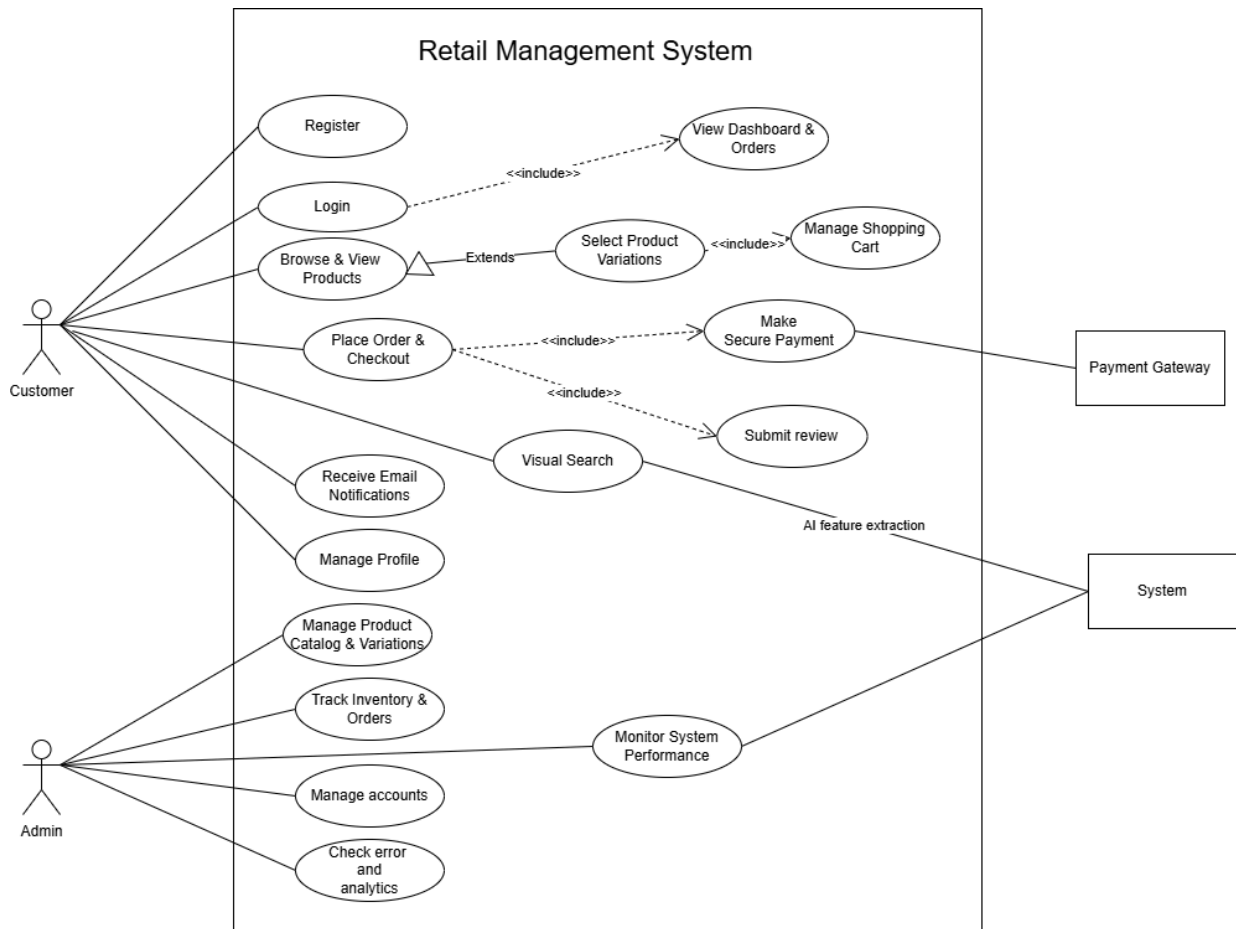


Fig. Use case Diagram

Non-Functional Requirements

Non-functional requirements describe the quality attributes, constraints, and performance expectations of the system. These requirements define how well the system should operate rather than what it does.

1. Usability

The system will be designed to be user-friendly and intuitive, allowing users with basic computer and internet knowledge to interact with the system easily without requiring extensive training.

2. Performance

The system is expected to provide acceptable response times during user interactions such as browsing, searching, and transaction processing to ensure a smooth user experience.

3. **Security**

The system will be designed to protect user data, transaction details, and system resources through secure authentication, authorization, and data handling mechanisms.

4. **Scalability**

The system should be capable of handling future growth in users, products, and transactions without significant changes to its core architecture.

5. **Reliability**

The system is expected to operate consistently with minimal downtime and should maintain data integrity even under unexpected conditions.

6. **Maintainability**

The system will be designed in a modular manner to simplify maintenance, debugging, and future enhancements.

7. **Compatibility**

The system should function correctly across common web browsers and devices to ensure broad accessibility for users.

4.2 Feasibility Study

The feasibility study evaluates the practicality and viability of the proposed Retail Management System from multiple perspectives to ensure that the project can be successfully developed and implemented within the given constraints.

4.2.1 Technical Feasibility

The proposed system is technically feasible as it will be developed using well-established and widely supported technologies such as HTML, CSS, JavaScript, jQuery, Bootstrap for the frontend, and Django with PostgreSQL for the backend and database management. These technologies are stable, open-source, and suitable for developing scalable and secure web-based applications. The development team has sufficient knowledge and access to the

required tools, frameworks, and development environments, making the technical implementation achievable.

4.2.2 Economic Feasibility

The project is economically feasible because it relies primarily on open-source technologies, which significantly reduces development and licensing costs. No expensive hardware or proprietary software is required for development or deployment. The system can be developed using standard computing resources, making it cost-effective and suitable for an academic project with limited financial constraints.

4.2.3 Legal Feasibility

The proposed system is legally feasible as it does not violate any known legal or regulatory requirements. The technologies and frameworks used are open-source and legally permissible for academic and educational use. The system will be designed to handle user data responsibly, following general data protection and privacy principles, ensuring compliance with applicable legal standards.

4.2.4 Operational Feasibility

The system is operationally feasible as it is designed to be user-friendly and easy to operate for both customers and administrators. The web-based nature of the system allows users to access it through standard web browsers without the need for specialized training. This ensures smooth adoption and effective day-to-day operation once the system is deployed.

4.2.5 Schedule Feasibility

The project is considered schedule feasible as it can be developed within the allocated academic timeline by following an Agile development approach. The use of modular development and iterative planning allows tasks to be managed efficiently, reducing the risk of delays and ensuring timely completion of the project milestones. For our project, we are using Gantt Chart.

Given below is a Gantt Chart describing the schedule for this Project:

Key activities	Duration (in weeks)														
	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th	13th	14th	15th
Planning															
Analysis & design															
Implementation															
Testing															
Evaluation															
Documentation															

Fig. Gantt chart

4.3 High Level Design

4.3.1 Methodology of the proposed system

The proposed Retail Management System will be developed using an Object-Oriented Design (OOD) approach combined with the Agile development methodology. The object-oriented approach is selected because it supports modular development, code reusability, scalability, and easier system maintenance. The system will be designed by dividing it into logical components where each component represents a specific entity of the system and manages its own data and operations.

4.3.2 Working Mechanism of Proposed System

Basic workflow of retail management system:

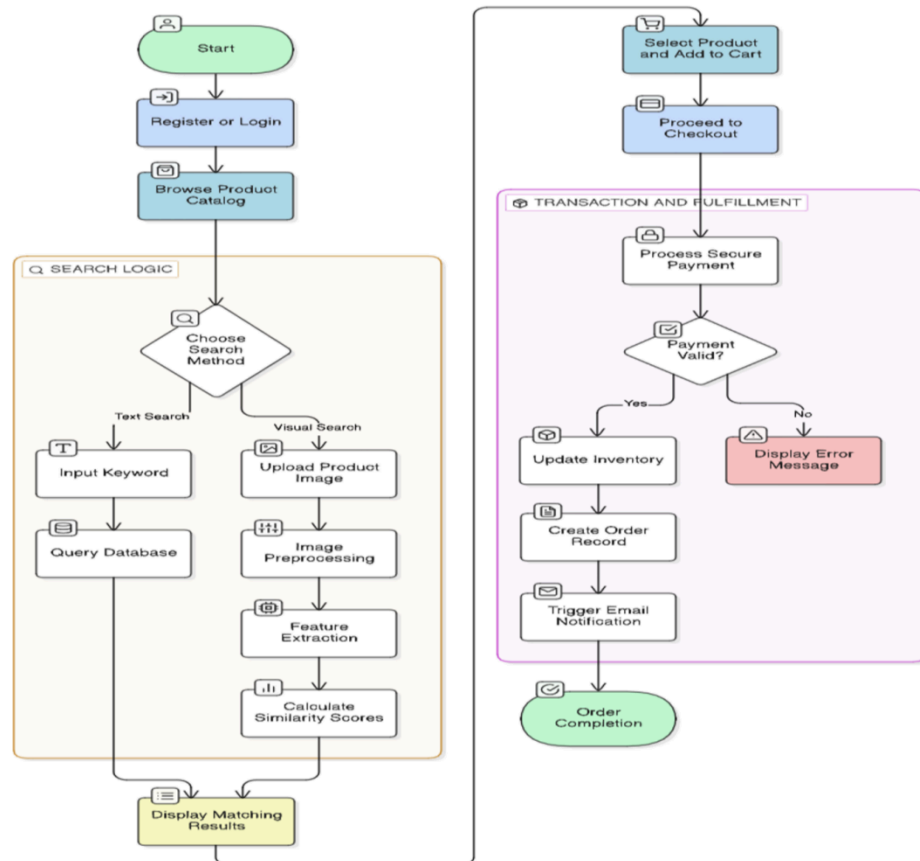


fig.4. Flowchart Diagram

4.4 Description of Algorithms

This section explains the core algorithms implemented in the Retail Management System, including visual similarity search, feature extraction, and review authorization. Each algorithm is described in terms of objectives, logical flow, and underlying mathematical logic.

Similarity Search Algorithm (Nearest Neighbors)

Objective:

- To identify and recommend products visually similar to a user-uploaded image.
- Enhance product discovery and improve user experience.

Logical Flow:

1. User uploads an image to the system.
2. The system extracts the image feature vector (2048-dimensional) using ResNet50.
3. The query vector is compared with all stored product vectors in the database.
4. Products are ranked by similarity based on distance metrics.
5. Top **N closest products** (e.g., 5) are returned as recommendations.

Mathematical Logic:

Euclidean Distance is used to measure similarity:

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

Feature Extraction Algorithm

Objective:

- To convert images into numerical representations (vectors) that capture their visual characteristics.
- Enable comparison between images in a high-dimensional space.

Logical Flow:

1. Pre-trained **ResNet50 CNN** is imported
2. Input images are resized to 224×224×3 and preprocessed.
3. Images are passed through the network to generate **feature vectors**.

4. Each image is represented as a **2048-dimensional vector embedding** stored in the database.
-

Review Authorization Algorithm

Objective:

- To ensure that only genuine buyers can submit product reviews.
- Prevent fake ratings and review manipulation, maintaining data integrity.

Logical Flow:

1. User attempts to submit a review for a product.
2. System checks the number of completed purchases CCC of the product by the user.
3. If $C \geq 1$, the review is allowed; otherwise, it is rejected.

Mathematical Logic:

Let:

U = User

C = Number of completed purchases by user for product

Allow Review = $\begin{cases} 1, & \text{if } C \geq 1 \\ 0, & \text{if } C = 0 \end{cases}$

5. Expected Outcome

The proposed Retail Management System is expected to deliver a functional, user-friendly, and efficient e-commerce platform that supports modern online retail operations through a centralized web-based application for managing products, users, orders, and transactions. A key expected outcome is enhanced product discovery through the integration of a visual search feature, allowing users to search for products using images instead of relying solely on text-based queries, thereby improving convenience, reducing search time, increasing user engagement, and enhancing the overall shopping experience.

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